

Use case name:	UC-1: Onboarding a New Lead	
Scenario:	A Real Estate Agent receives a message from a prospective client on social media and manually inputs it into REMS to parse client details, highlight uncertain information, and approve an automated follow-up reminder.	
Triggering event:	The agent receives an inquiry or message from a prospective client on a social media platform.	
Brief description:	This use case details how an agent transfers raw communication data from social media into REMS. The system auto-parses the text, highlights fields requiring validation, allows the agent to manually review, correct, or flesh out details, and sets up a profile and follow-up track on the system dashboard.	
Actors:	Real Estate Agent	
Related use cases:	- Includes: UC-1.3, UC-1.4, UC-1.5, UC-1.10 - Extends: UC-1.6 (extended by UC-1.5)	
Stakeholders:	Real Estate Agent, Prospective Client, Real Estate Agency Management	
Preconditions:	The Real Estate Agent is securely logged into the REMS application and has access to the New Client Intake Screen.	
Postconditions:	A new, active client profile is successfully saved to the database, their verified contact details are stored, and a follow-up reminder suggestion is approved and rendered on the agent's dashboard.	
Flow of activities:	Actor	System
	1. Receives a client message on social media (UC-1.1). 2. Pastes the message text into the New Client Intake Screen (UC-1.2). 3. Reviews and corrects the automatically extracted data fields (UC-1.5). 4. [Extension] Manually fills in the remaining Client Name &	1. Displays the New Client Intake Screen (UC-1.2). 2. Auto-parses message text and populates data fields (UC-1.3). 3. Highlights fields requiring validation or correction (UC-1.4). 4. Validates contact format entries (UC-1.6). 5. Saves the source tag to the client record (UC-1.7).

	Contact Details if needed (UC-1.6). - Confirms the source of the lead (UC-1.7). - Tags the client status as "Active" (UC-1.8). - Approves the system-suggested follow-up reminder timeframe (UC-1.9).	- Updates client status to Active in the database (UC-1.8). - Generates and displays a follow-up reminder suggestion on the dashboard (UC-1.9). - Saves the approved reminder and finalises the client profile (UC-1.10).
Exception conditions:	- System fails to parse text (UC-1.3): If the message text is corrupted or unreadable, the system displays an error message and requires the agent to manually complete all fields via UC-1.6. - Invalid contact format (UC-1.6): If the entered phone number or email fails formatting validation checks, the system rejects the submission until corrected.	

Figure 1: Fully developed use case description for “UC-1: Onboarding a New Lead”

Use case name:	UC-2: Property Status Update in the Field	
Scenario:	An agent logs a client's real-time feedback during a property viewing using a mobile device, triggering status updates, alternative property recommendations, and automated messaging drafts.	
Triggering event:	The agent completes or is actively conducting a property showing with a client out in the field.	
Brief description:	Allows mobile-optimised logging of showing results directly into REMS. Captures showing outcomes, updates listing status histories, recommends alternative properties contextually, and builds pre-formatted WhatsApp response strings for immediate communication.	
Actors:	Real Estate Agent	
Related use cases:	Includes: UC-2.2, UC-2.3, UC-2.5, UC-2.6, UC-2.7, UC-2.9	
Stakeholders:	Real Estate Agent, Client, Property Owner	
Preconditions:	The agent is logged into the REMS mobile interface and has an open, scheduled viewing session record active.	
Postconditions:	The viewing status history updates, property metrics adapt, and the agent is armed with an alternative listing text draft sent or ready to send to the client.	
Flow of activities:	Actor	System
	1. Opens the specific viewing record on their mobile device (UC-2.1). 2. Selects the explicit categorical outcome from the client feedback options (UC-2.3). 3. [Extension] Opts to add detailed custom feedback narrative notes (UC-2.4). 4. Edits, refines, and executes sending the generated message directly via REMS integrations (UC-2.8).	1. Loads the active viewing record and displays feedback options (UC-2.2). 2. Records the selected outcome and updates the showing status history (UC-2.3). 3. Saves the additional custom narrative notes if provided (UC-2.4). 4. Updates property listing metrics based on the recorded outcome (UC-2.5). 5. Searches for alternative properties matching the client's saved criteria (UC-2.6).

		6. Drafts a pre-formatted WhatsApp message with alternative property options (UC-2.7). 7. Dispatches the approved message via REMS integration (UC-2.8). 8. Logs the sent communication event to the client timeline (UC-2.9).
Exception conditions:	• No network connectivity: Mobile device loses data connection mid-field update. REMS caches the feedback locally and queues the synchronisation routine until connection returns. • No matches found (UC-2.6): System cannot find matching alternative parameters. It skips drafting the alternative properties snippet and logs a placeholder to alert the agent to look manually.	

Figure 2: Fully developed use case description for “UC-2: Property Status Update in the Field”

Use case name:	UC-3: Scheduling and Managing a Viewing	
Scenario:	An agent checks calendar availability via REMS, books a viewing time slot, synchronises it to Google Calendar, drafts a confirmation text, and subsequently updates status data post-viewing.	
Triggering event:	A lead requests a physical walkthrough appointment for a selected property listing.	
Brief description:	Handles end-to-end event setup for individual client property inspections by interfacing directly with third-party calendaring frameworks and text routing engines to minimise administrative friction.	
Actors:	Real Estate Agent, Google Calendar (External System Service Actor)	
Related use cases:	- Includes: UC-3.2, UC-3.4, UC-3.5, UC-3.6 - Extends: UC-3.2, UC-3.8 (extended by UC-3.7)	
Stakeholders:	Real Estate Agent, Buyer Client, Property Owner, External Calendar Provider	
Preconditions:	The relevant client account profile and property asset listing record must exist inside REMS. The agent's external Google account must be integrated.	
Postconditions:	The appointment is locked in REMS, mirrored to the external calendar asset, a WhatsApp reminder dispatch clears, and downstream pipeline paths open.	
Flow of activities:	Actor	System
	1. Navigates to the schedule viewing panel within REMS (UC-3.1). 2. Configures and selects an empty time slot, then submits appointment validation (UC-3.3). 3. Reviews, edits, and approves sending the confirmation via WhatsApp interface (UC-3.5). 4. Conducts viewing, then logs post-viewing feedback details and freeform narrative notes (UC-3.7).	1. Queries and displays available calendar slots via Google Calendar API (UC-3.2). 2. Validates the selected time slot and creates the appointment record in REMS (UC-3.4). 3. Synchronises the appointment to the agent's Google Calendar (UC-3.4). 4. Generates a pre-formatted WhatsApp confirmation message for the client (UC-3.5). 5. Dispatches the approved confirmation message (UC-3.6).

		6. Saves the post-viewing feedback and updates the listing status history (UC-3.8).
Exception conditions:	- Google Calendar API failure (UC-3.2 / UC-3.4): If the external calendar service times out or authentication drops, REMS falls back to internal scheduling logs, warns the user, and schedules a retry sync thread. - WhatsApp client unavailable (UC-3.5): If the target destination phone configuration lacks valid application hooks, the system defaults to generating an SMS fallback text layout.	

Figure 3: Fully developed use case description for “UC-3: Scheduling and Managing a Viewing”

Use case name:	UC-4: Managing a Deal Through Negotiation	
Scenario:	Multiple deal parties (Agent, Owner, Legal Rep) interact via REMS to submit offers, counter-offers, stage documents, sign final sales agreements, and finalise commissions.	
Triggering event:	An active buyer formally submits an initial monetary offer to purchase an inventory listing.	
Brief description:	A highly collaborative use case governing state flow across the formal legal binding and financial negotiation timeline of an active real estate transaction deal folder.	
Actors:	Real Estate Agent, Property Owner, Legal Representative	
Related use cases:	- Includes: UC-4.2, UC-4.4, UC-4.5, UC-4.10 - Extends: UC-4.7 (extended by UC-4.4)	
Stakeholders:	Buyer, Property Owner, Selling Agent, Legal Counsel, Escrow/Agency Financial Desk	
Preconditions:	The property asset must be marked active, and the client must possess an established deal pipeline record block.	
Postconditions:	Negotiation tracking updates, executing documentation uploads successfully finalise, deal records transition to closed-won status metrics, and agent commissions enter escrow state.	
Flow of activities:	Actor	System
	1. Real Estate Agent instantiates a new transaction track by choosing "Create Deal Record" (UC-4.1). 2. Real Estate Agent records the initial formal purchasing price bid (UC-4.2). 3. Property Owner signs in or communicates terms to log a counter-offer amount inside the portal tracker (UC-4.3).	1. Creates and initialises the deal record in the pipeline (UC-4.1). 2. Logs the initial offer and updates the deal tracker with the bid amount (UC-4.2). 3. Records the counter-offer and notifies the agent (UC-4.4). 4. Verifies and registers the Legal Representative's credentials (UC-4.6). 5. Validates and attaches uploaded compliance

	4. Legal Representative joins the transaction loop, providing formal legal representative verification metadata (UC-4.6). 5. Real Estate Agent uploads required milestone phase compliance documentation (UC-4.5). 6. Real Estate Agent marks negotiation complete and moves to "Advance to Final Sale Signed" (UC-4.8). 7. Real Estate Agent uploads the legally executed binding contract document artifact (UC-4.9).	documents to the deal record (UC-4.5). 6. Transitions the deal record state to Final Sale phase (UC-4.8). 7. Validates the uploaded contract, updates deal status to Closed-Won, and initiates commission escrow (UC-4.10).
Exception conditions:	• Document validation failure (UC-4.5): Uploaded PDF files are unreadable, unsigned, or corrupt. The system flags the item as "Rejected Compliance" and prevents the deal from moving to the signed phase. • Counter-offer rejection: Buyer rejects the owner's counter-offer out-of-band. The agent changes the deal record state to "Aborted Negotiation," which halts the active pipeline.	

Figure 4: Fully developed use case description for “UC-4: Managing a Deal Through Negotiation”

Use case name:	UC-5: Long-term Client Follow-up	
Scenario:	REMS surfaces a proactive alert matching a dormant client to a new listing; the agent reviews their history and delivers an AI-drafted text to sustain contact.	
Triggering event:	A long-term background system cron job or matching logic engine detects a new inventory item matching a past client's saved criteria, or a calendar reminder matures.	
Brief description:	Automates and streamlines contact routines for past or cold leads using automated notification triggers and AI message drafting to maintain high agency relationship touchpoints.	
Actors:	Real Estate Agent	
Related use cases:	- Includes: UC-5.3, UC-5.4, UC-5.6 - Extends: UC-5.2 (extended by UC-5.1)	
Stakeholders:	Real Estate Agent, Long-term Client, Marketing/Retention Director	
Preconditions:	The client profile exists in the system database with an "Inactive," "Past Buyer," or "Long-term Follow-up" categorisation tag.	
Postconditions:	A relationship touchpoint event logs to the database timeline, and a new forward-dated alert window is calculated and set.	
Flow of activities:	Actor	System
	1. Configures baseline temporal check-in rules or manually sets a checking parameter boundary (UC-5.1). 2. Interacts with the notification banner to select "View Client Profile & Interaction Summary" (UC-5.3). 3. Accesses the message review screen area to read, customise, or instantly execute the outbound dispatch (UC-5.4).	1. Monitors client profiles and triggers a notification when a matching listing is found or a reminder matures (UC-5.2). 2. Displays the client profile and full interaction summary (UC-5.3). 3. Generates an AI-drafted follow-up message tailored to the client's history and the matched listing (UC-5.4). 4. Dispatches the approved message and logs the communication touchpoint event (UC-5.5).

	4. Sends the finalised text message layout out to the client target destination (UC-5.5).	5. Calculates and schedules the next forward-dated follow-up alert window (UC-5.6).
Exception conditions:	• Opt-out constraint: If the client profile contains a "Do Not Contact" or privacy block flag, REMS suppresses all matching engine notifications and bars any outreach text generation routines.	

Figure 5: Fully developed use case description for “UC-5: Long-term Client Follow-up”

Use case name:	UC-6: Reviewing Monthly Finances	
Scenario:	An agent examines their monthly gross pipeline values, tracks individual property expenses, reviews net goals, and fires off payment reminders for missing commissions.	
Triggering event:	The agent chooses to check their sales performance metrics or requires business expense ledger entry updates.	
Brief description:	Provides comprehensive dashboard ledger reporting modules for calculating agent financial performance metrics, monitoring upcoming commission status trackers, and maintaining clean profit margins.	
Actors:	Real Estate Agent	
Related use cases:	- Includes: UC-6.2, UC-6.5, UC-6.6 - Extends: UC-6.3 (extended by UC-6.2)	
Stakeholders:	Real Estate Agent, Agency Accounting Office, Revenue Managers	
Preconditions:	Financial deal records, closed escrow balances, and expense logging fields must contain populated data within the active system environment.	
Postconditions:	Expense records are appended, financial summaries refresh, and pending commission alerts clear out-of-band to external targets.	
Flow of activities:	Actor	System
	1. Requests the primary financial accounting module inside the REMS cockpit and targets a specific month view (UC-6.1). 2. [Extension Condition] If a commission entry shows as past due, the agent can trigger an option to "Send Pending Commission Reminder" (UC-6.3). 3. Selects deal records to input specific operational out-of-pocket costs (UC-6.4).	1. Loads and displays the monthly financial dashboard with gross pipeline values, commissions, and expenses (UC-6.2). 2. Calculates and displays net goal progress against targets (UC-6.2). 3. Identifies past-due commission entries and highlights them on the dashboard (UC-6.3). 4. Sends the pending commission reminder to the relevant external target on agent approval (UC-6.3).

		5. Saves the entered expense records and refreshes the financial summary totals (UC-6.5). 6. Updates the expense ledger and recalculates profit margin data (UC-6.6).
Exception conditions:	• Missing tax/split configurations: If data profiles lack a defined agent commission percentage split model, calculation outputs default to standard agency baselines and append a warning banner requesting administrative review.	

Figure 6: Fully developed use case description for “UC-6: Reviewing Monthly Finances”

Use case name:	UC-7: Handling a Collapsed Deal	
Scenario:	A deal falls through; the agent marks it lost, and REMS automatically walks them through viewing recovery alternatives, picking properties, drafting a recovery text, and rescheduling a new follow-up.	
Triggering event:	A transaction in the negotiation or closing phase collapses completely (e.g., failed inspections, financing rejection, or client withdrawal).	
Brief description:	A tightly coupled linear workflow that guides an agent through the process of gracefully archiving a failed deal, analysing recovery alternatives, and transitioning the client to alternative inventory to keep the relationship alive.	
Actors:	Real Estate Agent	
Related use cases:	Includes: UC-7.2, UC-7.3, UC-7.4, UC-7.5, UC-7.6, UC-7.7	
Stakeholders:	Real Estate Agent, Buyer Client, Agency Management	
Preconditions:	An active deal record must be open in the negotiation or contract stage within REMS.	
Postconditions:	The original deal status shifts to "Closed - Lost," a historical loss reason is logged, alternative property targets are identified, a recovery message is dispatched, and a long-term follow-up task is scheduled.	
Flow of activities:	Actor	System
	1. Selects the active deal record and flags it as "At Risk," specifying the underlying collapse reason code (UC-7.1). 2. Reviews the suggestion matrix and chooses an appropriate pivot action path (UC-7.4). 3. Selects top candidate listings and moves to "Edit & Send Recovery Message to Client" (UC-7.6).	1. Updates the deal record status to "Closed - Lost" and logs the collapse reason to the history (UC-7.2). 2. Triggers the deal collapse workflow and notifies relevant stakeholders (UC-7.3). 3. Analyses the client profile and generates a recovery action suggestion matrix (UC-7.4). 4. Searches inventory for alternative properties matching the client's saved criteria (UC-7.5).

		5. Generates a pre-formatted recovery message draft with the selected alternative listings (UC-7.6). 6. Dispatches the approved recovery message and schedules a long-term follow-up task (UC-7.7).
Exception conditions:	No alternative inventory available (UC-7.5): If no matching inventory fits the client profile, the system skips the property review screen and prompts the agent to schedule an immediate discovery phone call via UC-7.7.	

Figure 7: Fully developed use case description for “UC-7: Handling a Collapsed Deal”